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# Interfacial and mechanical properties of liquid crystalline elastomer nanocomposites with grafted Au nanoparticles: A molecular dynamics study

Hongdeok Kim [a,b], Joonmyung Choi [a,b,*]

[a] Department of Mechanical Design Engineering, Hanyang University, 222 Wangsimni-ro, Seongdong-gu, Seoul, 04763, South Korea
[b] Department of Mechanical Engineering, BK21 FOUR ERICA-ACE Center, Hanyang University, 55 Hanyangdaehak-ro, Sangnok-gu Ansan, 15588, Republic of Korea

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ABSTRACT

Synthesizing surface-grafted Au nanoparticles (grafted-AuNPs) for incorporation into a liquid crystalline elastomer (LCE) matrix yields high contents of nanoparticles that are chemically compatible and soluble in the matrix. To investigate the change in the internal microstructure by inserting grafted-AuNPs and the mechanical role of the interfaces in the LCE/Au nanocomposites, all-atom molecular dynamics simulations were conducted. The results suggest that the insertion of the grafted-AuNP disrupts the LCE mesogen arrangement, especially in the interfacial area. Additionally, the strain energy density distributed to each molecular component revealed that the grafted unit on the side of the LCE matrix forms a highly entangled network with a semi-rigid microstructure and enables high load transfer efficiency. By contrast, a ductile grafted layer formed on the side of the AuNP surface acts as a cushion, which allows the AuNP to exist in the unloading state. Furthermore, the curvature of the inserted AuNPs is crucial in changing the mechanical and structural properties of the LCE matrix phase.

## 1. Introduction

The liquid crystalline elastomer (LCE) is a promising polymer that exhibits anisotropic properties according to the alignment of mesogen chains. Above a certain temperature, the mesogen units that are preferentially aligned in a particular direction lose their alignment at the molecular level, which leads to a contraction of the entire structure at the macroscopic level. Over the past decade, the actuation system of LCE spontaneously operating under contactless and powerless conditions has shown great potential as stimuli-responsive and functional materials. Because the deformation path and contraction force of the LCE can be controlled by both the direction and orientational order of the mesogen [57], LCEs have been developed for a wide range of applications, including artificial muscles [1–4], smart surfaces [5,6], microwave detector systems [55], and soft actuators [7–11]. Furthermore, mechanical programming techniques such as stretching [12–14], direct ink writing [4], imprinting [15], and 3D printing [6,8,9,11] diversify the actuation mode of the LCE with high precision.

To extend the versatility of LCE materials, there have been several attempts to produce multifunctional materials that respond to various triggers other than heating [54,55]. By incorporating nanoparticles into the LCE, various types of multi-stimuli-responsive actuators driven by light [16–21], magnetic fields [22], and electric-mechanical stimuli [23] are available. Together with additional functionalities, the presence of nanoparticles in the LCE also effectively dissipates locally generated heat with better thermal conductivity, thus improving the response speed and stability of the system [24].

However, poor chemical compatibility between the inorganic nanoparticles and the organic LCE matrix limits the maximum content of nanoparticles to only 1 wt% and make it challenging to obtain a stable nanocomposite phase with a high content of nanoparticles. As a strategy to improve the chemical compatibility, grafting organic molecules onto the nanoparticle surface was recently proposed [25,26]. This strategy increases the similarity between the nanoparticle and the organic matrix, and therefore, a very high content of nanoparticles within the LCE matrix can be achieved [26]. In addition, the organic molecules on the nanoparticle surface prevent undesired filler agglomeration that caused the deteriorated interaction with adjacent polymer matrix by increasing

* Corresponding author. Department of Mechanical Engineering, BK21 FOUR ERICA-ACE Center, Hanyang University, 55 Hanyangdaehak-ro, Sangnok-gu Ansan, 15588, Republic of Korea.
*E-mail address:* joonchoi@hanyang.ac.kr (J. Choi).

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the spatial distance between the nanoparticles [58]. Consequently, it has attracted considerable interest for the production of homogeneous dispersions of nanoparticles in the LCE matrix.

Nanoparticles with a high grafting density have spherical shell structures composed of grafted molecules. Therefore, two interfacial regions, nanoparticle/grafted molecules and grafted molecules/organic matrix, are present in the nanocomposites. Experimental and computational results show that the interface formed between the grafted unit and the organic matrix has a significant effect on the thermo-mechanical properties of the entire system. Pihan et al. claimed that the formation of a liquid-like layer at the interface between the matrix and the brush surface of silsesquioxane-based nanoparticles is a factor that reduces not only the glass transition temperature but also the elastic modulus [27]. Gao et al. observed that polyamide nanocomposites with highly grafted carbon nanotubes comprising very short grafted lengths exhibited a higher thermal resistance than an established prediction model [28], which was attributed to the new interface created by the grafted and free chains and its own thermal resistance. These studies indicate that the role of the interface between the grafted molecules and the adjacent matrix must be considered for the robust design of nanocomposites from a mechanical engineering perspective.

Another crucial aspect in the mechanical design of LCE nanocomposites is to capture the stress transfer capability through the interface. The mechanical properties of polymer nanocomposites are mainly determined by the efficiency of load transfer from the matrix to the nanoparticles [29]. In the case of nanocomposites using grafted nanoparticles in which two interfaces with different characteristics co-exist, the role of each interface in the reinforcement or degradation mechanism of polymer nanocomposites should be elucidated clearly. However, most of the experimental and computational studies so far have focused only on the macroscopic behavior of nanocomposites, and only a few studies have explored the role of each constituent of LCE nanocomposites and the behavior of these constituents at the interface.

Accordingly, in this study, the mechanical and interfacial characteristics of LCE nanocomposites with embedded surface-grafted Au nanoparticles (grafted-AuNPs) were investigated through all-atom molecular dynamics (MD) simulations. To ensure the reliability of the simulation model, the grafted-AuNPs in the LCE nanocomposites were prepared based on the experimentally reported composition [26]. In addition, by selecting different radii of AuNPs, the size effect on the inner morphology and the associated mechanical properties of the LCE nanocomposites were also characterized. To clarify the role of each interface in the material response under mechanical loading, the strain energy density allocated to the local region as well as each molecular component was quantified. This study provides a comprehensive understanding of the interfaces formed by grafted nanoparticles, which in turn facilitates the robust mechanical design of functionalized LCE nanocomposites.

## 2. Simulation models

In order to prepare nanoparticle surfaces that are covered with organic functional groups, thiolated Au nanoparticles were considered. In the grafted chain terminated with the thiol group (R-S-H), the S atom has a high affinity for the Au surface and is immobilized by forming covalent bonds with the nanoparticle surface. The interaction between Au and S comprises two distinct bond formations at the interface: the rectangular (Au [surf]-S [surf]-Au [lig]-S [surf]-Au [surf]) and V-shape (Au [surf]-S [surf]-Au [lig]-S [tip]-Au [lig]-S [surf]-Au [surf]) bonds, as shown in Fig. 1(a) [30,38]. Because the bond formation, type, and stability on the surface strongly rely on the surface geometry of the clusters, it is important to accurately determine the stable positions to locate the S and Au atoms. Regarding this issue, over the past 10 years, many researchers have revealed the natural deterministic positions of Au and S atoms on thiolated Au nanoparticles through X-ray diffraction, scanning tunneling microscopy (STM), and *ab initio* calculations [31–37]. Based

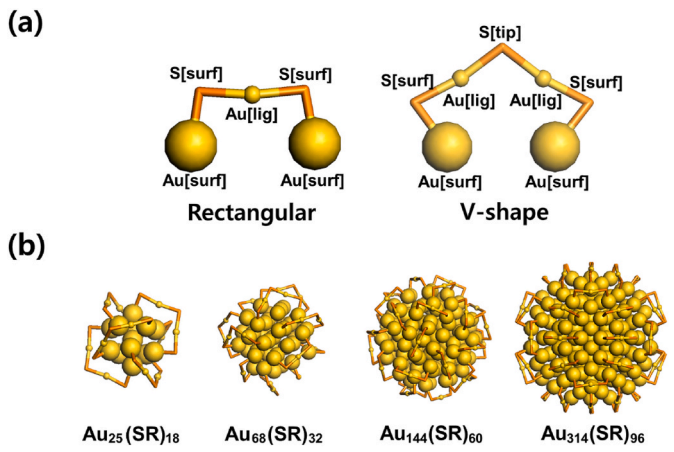



**Fig. 1.** Molecular configurations of thiolate-protected Au nanoparticles. (a) The two types of bond formation (rectangular or V-shape) produced by Au-thiol interactions and (b) thiolated Au nanoparticles of different sizes modeled in this study. The rectangular formation was found in Au68(SR)32, Au144(SR)60, and Au314(SR)96, while the V-shape formation was found in Au25(SR)18 and Au68(SR)32.

on these previous findings, we selected four types of nanoparticles to numerically investigate the effect of nanoparticle curvature; they are Au25(SR)18, Au68(SR)32, Au144(SR)60, and Au314(SR)96 (with a diameter of 1.2–2.3 nm), as illustrated in Fig. 1(b).

To describe the interatomic interactions of the pristine LCE and LCE/Au nanocomposite models, a polymer consistent force field (PCFF) was used in this study [39]. Since the PCFF does not contain the parameters to bind the Au and thiol groups, additional force field parameters of stretching and bending were added in order to consider the Au–thiol interaction. The force field parameters were obtained from a study by Pojlainen et al. [38]. They provide one set of parameters that can be applied to a wide range of clusters by simultaneously considering five representative thiolated Au nanoclusters with different diameters and types of ligands. The details are listed in **Table ST1**.

To evaluate the mechanical characteristics of transversely isotropic nanocomposites embedded with chemically modified nanoparticles, LCE and LCE/Au nanocomposites were prepared in all-atom MD simulations. The constituents of the microstructure used in this study are shown in Fig. 2. We selected the liquid crystal molecule 4-heptyl-4-cyanobiphenyl (7CB) and the backbone polymethylhydrosiloxane (PMHS) to model the nematic side-chain liquid crystalline elastomer.

With these components, two distinct types of crosslinkers were considered. The first is the linear diacrylate crosslinker used to form conventional polymer networks. The single diacrylate crosslinking agent generates a single bridge by covalently connecting the two neighboring PMHS backbones in the pristine LCE model. This provides a memory effect that maintains the programmed geometry as well as improves the thermal stability of the LCE system [56]. Initially, the liquid crystal monomers, PMHS backbones, and diacrylate crosslinking agents were randomly distributed in a cubic unit cell. The details of the internal composition of all the LCE and LCE/Au models are presented in Table 1. Periodic boundary conditions were applied to all directions to simulate a continuous bulk material that interacts across boundaries. The mesogens of the liquid crystal monomers preserved their orientation along the x direction, forming a transversely isotropic phase. The constituent molecules of the system were located in a low-density environment of 0.2 g/cc to prevent unintended interference between the molecules. Then, each model was gradually compressed until the density of the unit cell reached 1.0 g/cc while preserving the alignment of the mesogen units. For each step of compression, the internal potential energy of the microstructure was minimized using a conjugated gradient method with a cutoff value of 10 kcal/mol Å. After energy minimization, dynamic

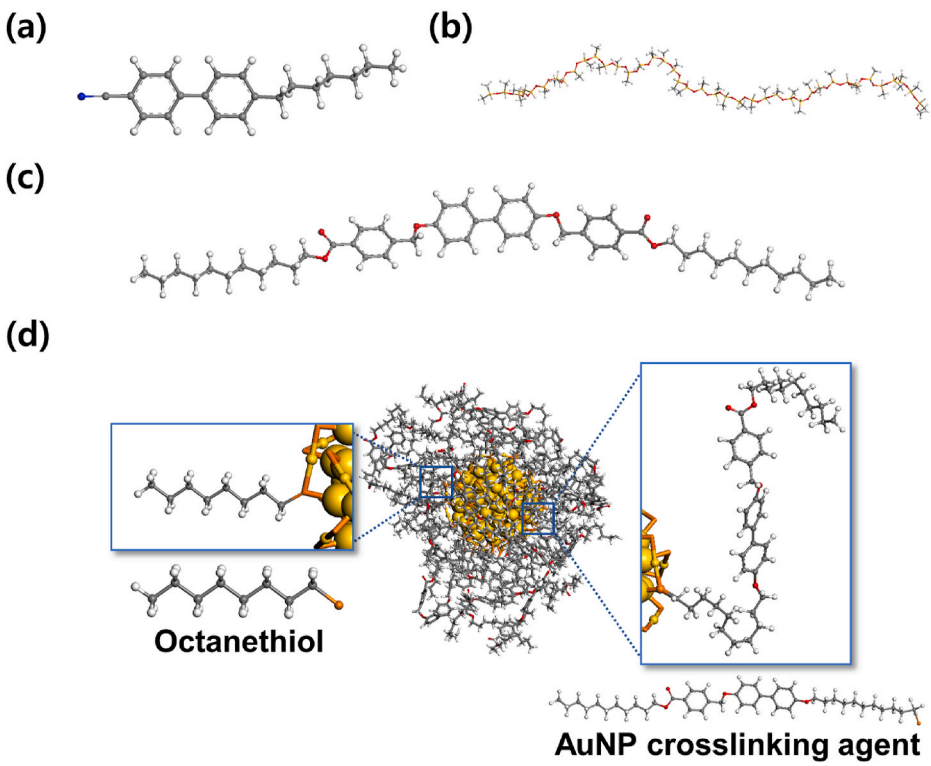



**Fig. 2.** Chemical structures of the components of LCE and LCE/Au nanocomposites modeled in this study. (a) The liquid crystal monomer 4-heptyl-4′-cyanobiphenyl (7CB), (b) the polymethylhydrosiloxane (PMHS) backbone, (c) the diacrylate crosslinking agent, and (d) the Au144 (SR)60 gold nanoparticle grafted with octanethiol (70% of surface sulfur) and AuNP crosslinking agent (30% of surface sulfur). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

**Table 1**
Composition of the LCE and LCE/Au nanocomposites.

| | Molecular weight ($gmol^{-1}$) | Number of molecule chains | | | | |
|---|---|---|---|---|---|---|
| | | Neat LCE | LCE/ Au25 | LCE/ Au68 | LCE/ Au144 | LCE/ Au314 |
| Grafted-AuNP | – | – | 1 | 1 | 1 | 1 |
| Diacrylate crosslinking agent | 763.07 | 2 | – | – | – | – |
| PMHS backbone | 2435.15 | 7 | 4 | 7 | 12 | 24 |
| LCE monomer (7CB) | 277.41 | 215 | 115 | 215 | 396 | 756 |

simulation was performed with the NVT ensemble at 300 K for 1 ns, followed by the NPT ensemble at 300 K and 0.1 MPa for 1 ns with a time step of 1 fs using the Verlet velocity algorithm.

The second is the Au nanoparticles grafted with organic groups as exclusive crosslinkers. The architecture of diacrylate crosslinking agents and grafted-AuNPs was constructed according to the report by Wójcik et al. [26]. In order to modify the surface of the AuNPs, 70% of the surface S atoms were bonded to octanethiol and the remainder were connected to the AuNP crosslinking agents. Before they were inserted into the LCE matrix, the modeled grafted-AuNPs were equilibrated in the NVT ensemble at 500 K for 500 ps with a 1 fs time step. Then, the LCE/Au nanocomposite networks were generated using the conditions of the neat LCE network model except that the centered AuNP crosslinking agents anchored the adjacent PMHS backbone members instead of the diacrylate members. It is a well-known fact that the effective material properties of nanocomposites are highly dependent on the size of the inclusion. To systematically examine the effect of the curvature of the AuNPs on the heterogeneous microstructure, four LCE/Au nanocomposites with differently sized AuNPs were modeled under a constant volume fraction of 1 vol%.

The crosslinking reaction for the uncrosslinked LCE and LCE/Au nanostructures was carried out as follows. The reaction process was separated into two steps: (1) the formation of the side-chain mesogenic polymers and (2) the interconnection of the networks via crosslinkers. In each step, the relative distance between the reactive sites of the components to be coupled was monitored, and artificial covalent bonds were newly formed when the distance between the reactive sites was shorter than the cutoff length. A flowchart of the entire crosslinking procedure is illustrated in Fig. 3(a). In the first step, the alkyl side chains of the liquid crystal monomers were connected to the surrounding PMHS backbone members to form the side-chain nematic LCE. The cutoff distance started at 4.0 Å and continuously increased with a step increment of 0.5 Å until the crosslinking ratio reached 40%. At each cutoff distance, geometry optimization using the conjugated gradient method with a cutoff value of 10 kcal/mol Å and a short relaxation consisting of heating from 300 K to 500 K and cooling from 500 K to 300 K for a total time of 30 ps was executed. After the first crosslinking reaction was completed, the LCE and LCE/Au nanostructures were equilibrated using the NVT ensemble at 300 K for 0.5 ns followed by the NPT ensemble at 300 K and 0.1 MPa for 0.5 ns. The second crosslinking step interconnected the diacrylate or AuNP crosslinking agents and the PMHS backbone chains to compose LCE networks. Similar to the first step, the distance between the reaction points of the crosslinking agents and the PMHS backbones was monitored and covalently bonded according to the cutoff distance. The second step was performed until all the diacrylates (or AuNP crosslinking agents) were completely immobilized on the PMHS backbones. The crosslinked LCE and LCE/Au nanocomposite models were further relaxed using the NVT ensemble at 300 K for 2 ns and the NPT ensemble at 300 K and 0.1 MPa for 3 ns. Note that the present method has been widely used to constitute cured polymer resins in MD simulations [40, 41] because it can suitably represent the intertwined morphology of the molecular assembly beyond the periodic boundaries as well as within the unit cell. The local network structures after each crosslinking reaction are depicted in Fig. 3(b).

Note that the length of the unit cells and the corresponding volume fraction of nanoparticles can be changed during isothermal-isobaric dynamics. These variations differed depending on the curvature of the nanoparticles and made a difference in the volume fraction of

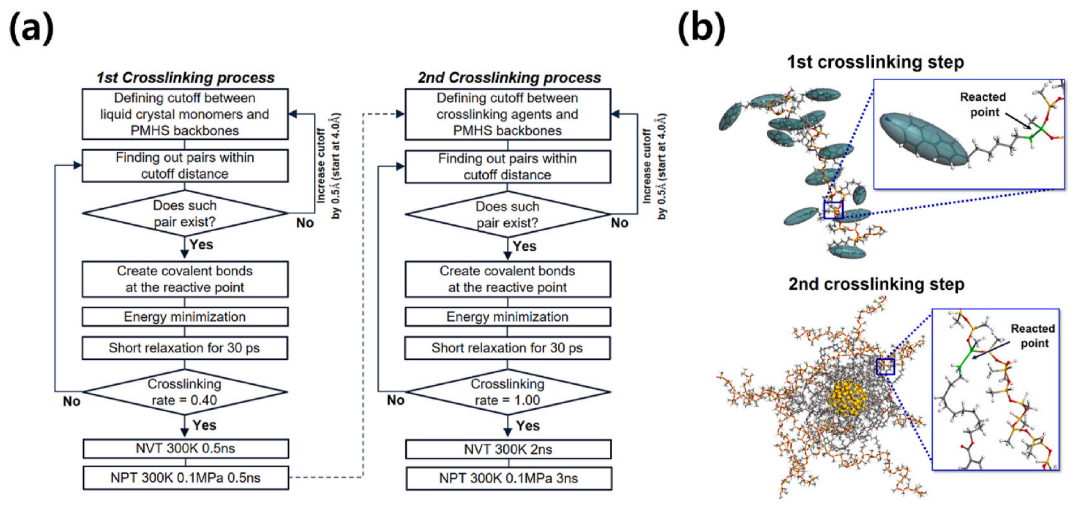



**Fig. 3.** Two-step crosslinking process for side-chain LCE and polymer networks. (a) A flowchart of the crosslinking procedure and (b) the local molecular structures after each crosslinking step.

approximately <0.5% when compared with the target volume fraction. In order to clarify the difference, the details of the LCE and LCE/Au nanocomposite models prepared in this study are shown in Table 2. The slight increase in the nanoparticle volume fraction (1.18–1.3%) was not critical enough to affect the material properties of the nanocomposites. Thus, the constructed models were still valid to observe independently the interfacial properties by the differently sized nanoparticles under constant composition conditions.

The LCE/Au nanocomposite models possessed a higher density than the neat LCE model. Moreover, the density of the LCE/Au nanocomposites further increased as the curvature of the nanoparticles was reduced. Generally, metallic or ceramic nanoparticles with a uniform composition have a constant density regardless of the size of the nanoparticle. On the other hand, the thiolated AuNPs used in this study were covered with sulfur, a relatively light element, only on the surface of the particles. This compositional feature in which the particles possessed an enormous surface-to-volume ratio was the major factor that determined the density of the nanocomposites. Accordingly, the smaller thiolate-AuNP formed a material with a lower density due to the high proportion of sulfur atoms.

The complete LCE and LCE/Au nanocomposite structures are shown in Fig. 4. During the crosslinking procedure and the series of relaxations, the mesogen units maintained alignment in the x direction and formed a nematic phase. In addition, the grafted-AuNP located at the center of the unit cell exhibited a highly entangled morphology in all directions with the adjacent LCE components. It was verified that the geometry of the thiolated AuNPs modeled by the force field parameters in Ref. [38] preserved its desired conformation in the LCE matrix, with good agreement with the reported values, as shown in Table 3.

The all-atom modeling of the LCE and LCE/Au nanocomposites was carried out using the commercial MD software Materials Studio (Dassault Systemes, 2020). The mechanical properties and the relevant microstructural characteristics were obtained through the open-source MD code, LAMMPS (Sandia Lab).

## 3. Results and discussion

### 3.1. Radial density distribution of LCE/Au nanocomposites

For each LCE/Au nanocomposite, the radial density distribution profiles of all the constituent molecules except AuNP were obtained and are shown in Fig. 5(a). The density profiles were extracted by dividing the unit cell into a spherical shell concentric with the AuNP (0.2 Å interval) and the local density within the shell region was calculated. The method can clearly observe the change in microstructure according to the adsorption properties formed by the nanoparticles and the surrounding polymer layer. In this regard, numerous studies have shown that the local density reaches a peak value near the nanoparticle surface owing to the strong non-bonded interaction between the nanoparticles and the adjacent polymer constituent [42–46]. However, the results of this study showed that such density distribution did not appear at all, but rather quickly converged with a uniform profile throughout the entire system.

The main reason for this non-trivial result is that the morphology change induced by the non-bonded interaction between the LCE components and AuNP was significantly reduced due to the grafted molecules onto the AuNP surface. As the grafted molecules occupy the interfacial region between the LCE components and AuNP, the LCE components increase the spatial distance from the AuNP, and thus the interface effect acts dominantly only on the grafted molecules. Fig. 5(b) shows the radial density profile of each of the grafted molecules and the LCE matrix. The grafted members covalently attached to the AuNP form a highly densified layer right next to the nanoparticle surface. Therefore, the corresponding density profile of the grafted molecules has a peak on

**Table 2**
LCE and LCE/Au nanocomposite systems modeled by MD simulations.

| Model | AuNP diameter (Å) | Targeted AuNP volume fraction (%) | AuNP volume faction after equilibration (%) | 1st step crosslinking rate (%) | Cubic unit cell length (Å) | Density (g/cc) |
|---|---|---|---|---|---|---|
| Neat LCE | – | – | – | 42.8 | 50.42 | 1.02 |
| LCE/Au25 | 12 | 1 | 1.18 | 44.3 | 42.53 | 1.12 |
| LCE/Au68 | 15 | | 1.24 | 44.7 | 52.19 | 1.16 |
| LCE/Au144 | 18.5 | | 1.28 | 41.4 | 63.81 | 1.18 |
| LCE/Au314 | 23 | | 1.3 | 41.8 | 78.94 | 1.21 |

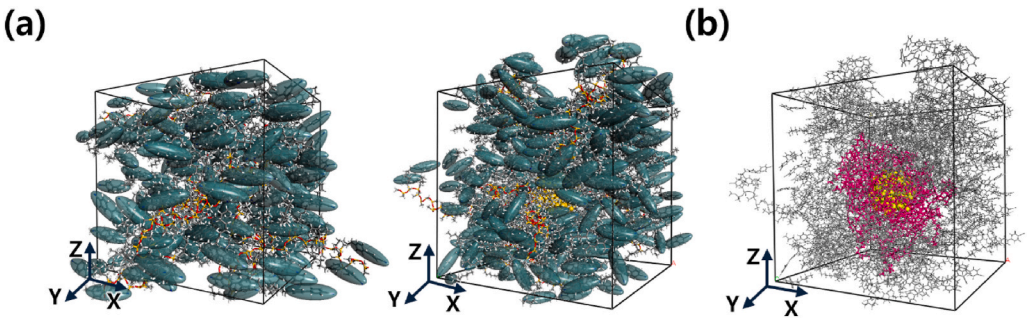



**Fig. 4.** (a) Crosslinked LCE (left) and LCE/Au nanocomposites (right) modeled in the MD simulations. The mesogen units are highlighted as the ellipsoidal inclusions. (b) The highlighted view of the AuNP and grafted molecules in the LCE/Au nanocomposites model.

**Table 3**
Averaged geometry parameters of the thiolated AuNP.

| | Bond length (Å) | | Angle (deg) | | | Pair distance (Å) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Au [surf]-S [surf] | Au [lig]-S [surf] Au [lig]-S [tip] | Au [surf]-S [surf]-Au [lig] | S [tip]-Au [lig]-S [surf] | Au [lig]-S [tip]-Au [lig] | Au [cen][a]-Au [surf] | Au [cen]-Au [int] | Au [cen]-S [surf] | Au [cen]-S [tip] |
| Au25 | 2.33 | 2.27 | 78.23 | 158.16 | 97.92 | 3 | 4.95 | 5.23 | 5.93 |
| Au68 | 2.36 | 2.27 | 85.61 | 158.54 | 90.08 | 5.36 | 6.72 | 7.1 | 7.94 |
| Au144 | 2.35 | 2.29 | 75.55 | 162.74 | N/A | 7.02 | 8.77 | 9.02 | N/A |
| Au314 | 2.43 | 2.35 | 78.91 | 161.56 | N/A | 9.54 | 11.6 | 11.88 | N/A |
| Force field equilibrium [38] | 2.41 | 2.33 | 91.3 | 172.4 | 100 | N/A | N/A | N/A | N/A |
| Au25[b] [38] | N/A | N/A | N/A | N/A | 99.01 | 2.87 | 4.85 | 4.97 | 6.09 |

[a] Au [cen] indicates the center of Au nanoparticles.
[b] In solvent condition.

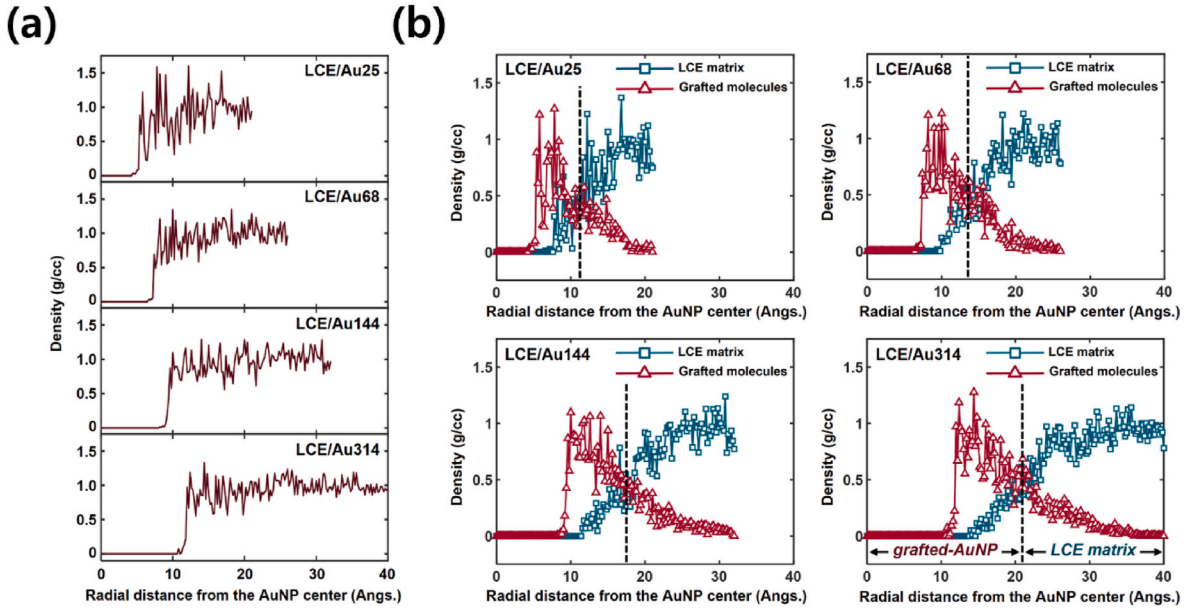



**Fig. 5.** Radial density distributions of LCE/Au nanocomposites after equilibration. The mass of AuNPs was subtracted from the measurement for clarity. The density profiles of (a) all the polymeric constituents and (b) the components separated into grafted molecules and LCE matrix.

the AuNP surface, and its distribution characteristics rapidly diminish with increasing distance from the surface. On the other hand, the density of the LCE matrix gradually increased as it moved away from the AuNP surface and monotonically converged at the same density as the neat LCE (1.0 g/cc). It is worth nothing that the grafted-AuNPs were equilibrated independently before being inserted into the LCE matrix to emulate the synthesis process [26]. Since a crowded grafted layer had already formed on the AuNP surface prior to interaction with the LCE components, all the LCE components, including the PMHS backbone chains, hardly penetrated the AuNP surface but were distributed only outside the grafted layer. Details of this density distribution of the LCE matrix by components can be found in Figure S1 in the Supplementary Data. Consequently, the intersection of the density profile of the grafted molecules and the LCE matrix components occurs 5 to 10 Å away from the nanoparticle surface, which indicates that the grafted layer causes a large separation between the AuNP and LCE components. The buffered interaction ensures that the geometric and structural properties of the LCE matrix within the nanocomposites are similar to those without

nanoparticles.

Obviously, it is difficult to clearly separate the phases of the grafted units and the LCE matrix because they are intricately entangled on the nanoparticle surface. To define the interface formed by the grafted units and the LCE matrix, we considered an imaginary spherical interface that is concentric with the AuNP and has a radius determined by the intersection point of each profile of the grafted-AuNP and the LCE matrix (see LCE/Au314 in Fig. 5(b)). The intersection distance from the AuNP center and the associated volume fraction of the grafted-AuNP in the unit cells are shown in Table 4. Although the AuNP was modeled to have a constant volume fraction of 1%, the volume fraction occupied by the AuNP and the grafted layer steadily increased from 7.45% to 8.76% as the size of the AuNPs increased. In other words, the structural conformation of the grafted units was significantly varied depending on the size of the nanoparticles.

### 3.2. Morphology of grafted units

To better understand the size dependence of the morphology of the grafted units on the AuNP surface, the radius of gyration (*Rg*) of octanethiol, the main components of the grafted molecules, was calculated. The distribution of the *Rg* of polymer chain is determined as follows:

$$< R_g^2(N) > = \frac{1}{N}\langle \sum_{i=1}^{N} (r_i - r_{cm})^2 \rangle \quad (1)$$

where $r_i$ and $r_{cm}$ are the position vectors of each carbon atom and the center-of-mass, respectively. As shown in Fig. 6(a), the *Rg* of octanethiol gradually increased as the size of AuNP increased, indicating that a well-ordered conformation was generated on the AuNP surface. Also, the average orientational angle ($\theta$) of the octanethiol carbon atoms supports this result. A virtual line segment penetrating the center of mass of AuNP and the center-of-mass of octanethiol was considered and the angle distribution between each carbon atom and the line segment was calculated. The result showed that octanethiol was narrowly distributed from the line segment for large nanoparticles (i.e., $\theta$ decreased as the curvature of AuNP decreased) and was more likely to be perpendicular to the AuNP surface.

We also derived the average distance from the sulfur atom to each carbon atom of the grafted unit, and the results are depicted in Fig. 6(b). As the particle size increased, a set of carbon atoms that comprised the grafted unit moved farther from the nanoparticle surface, and thus the grafted-AuNP occupied a larger volume fraction in the unit cell. This shows good consistency with the other results discussed above.

The relation between the structural features of the grafted molecules and the particle size is due to the variation in the grafting density and their curvature characteristics [46–48]. As the size of AuNP increases, the grafting density of thiolated AuNP, defined as the ratio of grafted chains per surface area, gradually increases (from 3.98 *chains*/ $nm^2$ to 5.78 *chains*/$nm^2$) and the space between the grafted chains is more limited due to a lower surface curvature of the nanoparticle [46]. Consequently, the grafted layer of the large nanoparticles becomes more congested, which forces the grafted molecules to form an extended conformation. Another factor that strongly influences the morphology of the grafted molecules is the non-bond interaction space of the nanoparticle surface atoms. For larger nanoparticles, the surface of each particle becomes relatively flat, providing spatially limited non-bond interactions between the grafted units and the surface atoms on the nanoparticle [41]. However, as the size of the nanoparticles decreases, the surface atoms acquire a broad interaction space with the grafted units owing to their high curvature, which induces a more curled and tilted morphology of the grafted molecules.

**Table 4**
Intersection distance and the volume fraction of grafted-AuNP.

| Model | LCE/ Au25 | LCE/ Au68 | LCE/ Au144 | LCE/ Au314 |
|---|---|---|---|---|
| Intersection distance from the AuNP center (Å) | 11.1 | 14.2 | 17.6 | 21.8 |
| Volume fraction of the grafted-AuNP (%) | 7.45 | 8.35 | 8.72 | 8.76 |

### 3.3. Molecular alignment of mesgoen in the LCE matrix

In order to observe the change in the alignment state of the LCE matrix by the insertion of the grafted-AuNP, the directionality of the mesogens was evaluated. The line segments connecting both ends of the mesogen were considered and then, the angle ($\theta$) between the considered line and the x-axis (nematic director) was measured. The alignment properties of the LCE were evaluated through the scalar orientational order parameter (*S*) that can be derived as follows:

$$S = \langle \frac{3cos^2\theta - 1}{2} \rangle \quad (2)$$

The value of *S* is close to unity if the constituents are perfectly aligned towards the reference axis, whereas the value approaches zero for a randomly distributed conformation. The alignment was measured radially with an interval of 1 Å from the intersection point derived in the above section. As a result, as shown in Fig. 7(b), there was significant variation in the scalar orientational order of the LCE matrix. In particular, the mesogen units considerably lost their initial alignment near the interface. This result is contradictory to our previous studies using nanoparticles without grafted units [40,49]. When the size of nanoparticles is on the order of the mesogen length, the mesogen arrangement is weakly disturbed only near the interface, and the initial orientation is well preserved throughout the entire system. However, the grafted-AuNPs used in this study have an enlarged effective size (ranging from 22.2 to 43.5 Å in diameter) due to the thick grafted layer, which is much larger than the characteristic length of the mesogen (~10.7 Å). Consequently, the LCE mesogens near the interface did not maintain the original orientation until the disrupted alignment gradually recovered as it moved away from the interface.

Another remarkable point is that the alignment properties of the LCE/Au nanocomposites were not sufficiently recovered to the level of the neat LCE even until the measurement area reached the boundary of the unit cell. When the size of the inserted particles decreases, the distribution of the orientation characteristics of the LCE matrix further deteriorates. This is plausible because the volume of the LCE matrix for alignment recovery was reduced as the size of the grafted-AuNP decreased. The grafted-AuNP modeled in this study formed evenly spaced patterns under the 3D periodic boundary conditions with high concentration, and the LCE matrix filled the region between these particles (see illustration in Fig. 7(a)). Accordingly, the whole matrix domain in the nanocomposites is influenced by the distortion, and the space for full restoration of the alignment is insufficient. In particular, as the size of AuNPs decreases, the absolute distance between adjacent grafted-AuNP surfaces further decreases (from 35.44 to 20.3 Å in the model considered), further reducing the spatial region of LCE for restoration of alignment. In this contracted LCE region, the effect of the grafted-AuNP does not diminish but acts to disrupt the alignment across the entire domain.

The orientational order of the 7CB modeled in this study and the value of pure nematic 7CB reported in previous computational and experimental studies [52,53] are shown in Table 5. Note that the models considered are fully equilibrated through a series of dynamics for each stage of relaxation. The convergence of the orientational order during MD simulation can be found in Figure S2. The orientational order of the neat LCE model agreed with the experimental observations, even though a slight underestimation was found. This is due to the randomly arranged components (i.e., PMHS backbones and crosslinkers), and the crosslinking reaction further interferes with the spontaneous arrangement of the LC monomers. For all the LCE/Au nanocomposite models

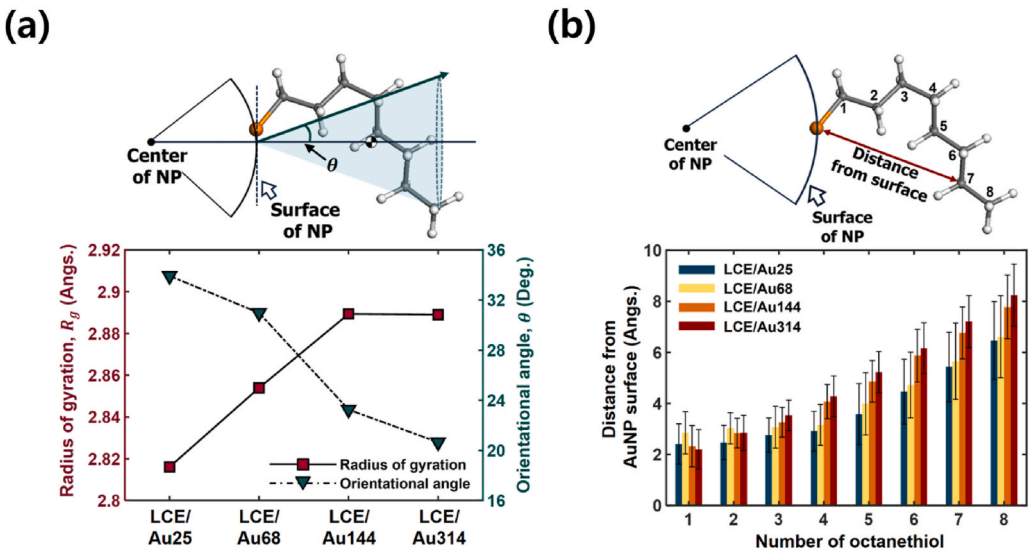



**Fig. 6.** Morphology analysis of the grafted molecule layer within the LCE/Au nanocomposites. (a) Radius of gyration (*Rg*) and orientational angle ($\theta$) of the octanethiol carbon atoms and (b) the distance of each carbon atom from the AuNP surface. Error bar represents the standard deviation.

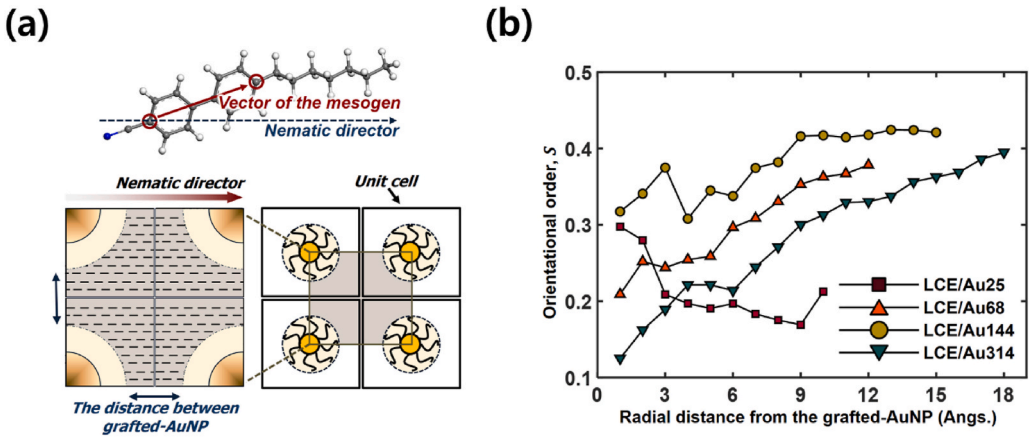



**Fig. 7.** Alignment characteristics of liquid crystalline molecules. (a) Schematic of the measurement of orientational order (top) and the periodic conditions of the unit cells (bottom) and (b) the radial distribution of the scalar orientational order according to the size of the AuNP.

**Table 5**
Orientational order of the modeled system.

| | Neat LCE | LCE/Au25 | LCE/Au68 | LCE/Au144 | LCE/Au314 | Refer [52]. | Refer [53]. |
|---|---|---|---|---|---|---|---|
| LC material | 4-heptyl-4′-cyanobiphenyl (7CB) | | | | | | |
| Crosslinking rate (%) | 42.8 | 44.3 | 44.7 | 41.4 | 41.8 | None | None |
| Orientational order (S) | 0.493 | 0.218 | 0.404 | 0.468 | 0.386 | 0.555 | 0.603 |

except for the one involving the smallest particle size, comparable orientational order characteristics were observed. The results also showed good agreement with the experimental study of Wójcik et al. [26] in which the crystallization of the LCE matrix was clearly observed despite the addition of 2.2 nm diameter AuNPs by 2 wt%.

### 3.4. Mechanical properties of the LCE and LCE/Au nanocomposites

The elastic mechanical properties were measured by uniaxial tensile simulation. All the LCE and LCE/Au nanocomposite models were relaxed at 0.1 K for 500 ps using the NVT ensemble and were elongated at an engineering strain rate of $10^6$ $s^{-1}$ until the maximum strain reached 0.25%. During the tensile loading simulation, the atomic stress tensor ($\boldsymbol{\sigma}$) was extracted based on the virial theorem that can be obtained as follows [50]:

$$\boldsymbol{\sigma} = \frac{1}{V}\left( -\sum_i^N m_i\left(\boldsymbol{v}_i\boldsymbol{v}_i^T\right) + \frac{1}{2}\sum_i^N\sum_{j\neq i}^N \boldsymbol{r}_{ij}\boldsymbol{F}_{ij}\right) \quad (3)$$

where V is the volume of the system, N is the total number of atoms, $m_i$ and $v_i$ are respectively the atomic mass and velocity of the $i$ -th atom. In addition, $r_{ij}$ and $F_{ij}$ denote the distance and the interatomic force between the $i$ -th and $j$ -th atoms, respectively. Under the extremely low temperature environment of 0.1 K, the kinetic term (the first term) of Eq. (3) is negligible. Therefore, the external force exerted on the microstructure during tensile deformation is directly converted into the interatomic potential term (the second term) of the constituent atoms, and $\boldsymbol{\sigma}$ can be regarded as the mechanical Cauchy stress [51]. The elastic

stiffness tensor ($\boldsymbol{C}$) of the microstructure can be derived as follows:

$$\boldsymbol{\sigma} = \boldsymbol{C\varepsilon} \tag{4}$$

where $\varepsilon$ is the strain applied to the unit cell. Additionally, to clarify the uncertainty induced by the initial configuration, all the mechanical tests were repeated two more times in which additional equilibrations were conducted (under the NPT ensemble at 300 K and 1 atm) for 50 ps and 100 ps, and the calculation results were averaged. In these cases, the maximum strain was limited to 0.05% and then extrapolated using linear prediction.

Fig. 8 shows the elastic stiffness in the axial and transverse directions and the scalar orientational order of the entire LCE matrix. Both the LCE and LCE/Au nanocomposite models have transversely isotropic properties due to the highly oriented LCE nanostructure along the nematic axis. In addition, the stiffness anisotropy between the axial and transverse directions varied in accordance with the value of the orientational order parameter. The LCE/Au25 nanocomposites, in which the directionality of the system was greatly disturbed because of the short distance between the grafted-AuNPs, exhibited a small difference in the axial and transverse stiffness and were more likely to behave as an isotropic material. On the other hand, for the other three nanocomposites, the orientation characteristics of the LCE matrix were greatly restored and the corresponding stiffness values were almost identical regardless of the size of the grafted-AuNP. Nevertheless, even though the orientational order is recovered to the level near that of the neat LCE, the stiffness components of the three nanocomposite models were inferior to those of the neat LCE in both directions.

It should be stressed that the present results do not conclude the adverse effects of inserting AuNPs on the mechanical properties. The grafting layer design applied to the present model was suggested only to maximize the loading of the AuNPs to the LCE matrix. Therefore, the variation of the overall mechanical stiffness other than the structural stability of the microstructural phases has not been sufficiently considered. In this study, we found that the mechanical role of the interfaces achieved by the grafting layer is key to determining the overall properties of the nanocomposite system, especially when the characteristic size of the grafting layer is comparable to or exceeds that of the AuNP. This will be further discussed in the next paragraph.

To examine the cause of the deteriorated mechanical performance of the LCE/Au nanocomposites, the local strain energy density around the nanoparticle was calculated after the model was subjected to tensile deformation. A thin spherical shell of increasing thickness with an interval of 1 Å was considered, and the strain energy density ($u$) at each thickness was derived as follows:

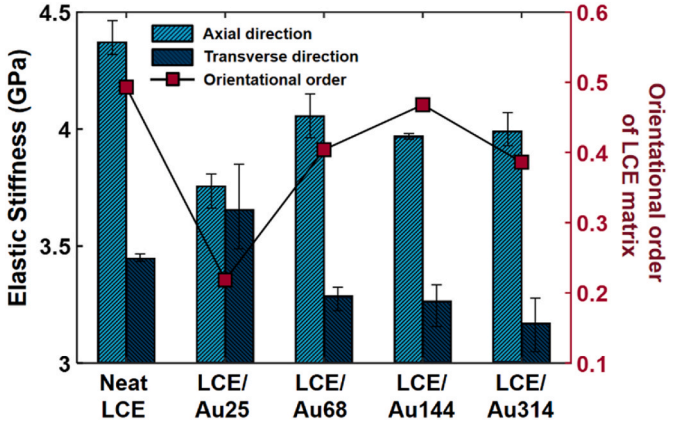



**Fig. 8.** Elastic stiffness in the axial and transverse direction and the orientational order of whole LCE matrix according to the size of AuNP. Error bar represents min-max range.

$$u = \int_0^{\varepsilon_0} \sigma \, d\varepsilon \tag{5}$$

where $\sigma$ and $\varepsilon$ are respectively the stress and strain applied to the spherical shell region and $\varepsilon_0$ denotes the maximum strain. Because the LCE/Au nanocomposite unit cells were under a constant displacement condition, the distribution of the strain field was determined by the mobility and deformability of the components in the region. The regions of the grafted layer and the LCE matrix were considered separately, and the profiles at each segmented region were obtained as shown in Fig. 9 (a) and Fig. 9(b), respectively.

In both loading directions, the highest strain energy density throughout the entire system was observed immediately next to the AuNP surface. This result suggests that the layer made by grafted units possesses considerable tensile ductility, especially on the surface of the AuNPs. The linear structure of the grafted chain adaptively responds to the external mechanical load and is easily deformable via inter- and intra-molecular interactions. However, the grafted chain gradually interlocks with the LCE matrix as the distance from the AuNP surface increases and the corresponding entangled morphology improves the structural rigidity to resist mechanical strain. As a result, deformation was dominant only near the AuNP surface, which gradually reduced as the measurement point moved away from the AuNP surface. In other words, the external load applied to the grafted layer barely transferred to the rigid AuNP. Meanwhile, the strain energy density profiles of the LCE matrix converged to low levels without the appearance of significant distribution features in both loading directions. Even though the LCE mesogens significantly lost their initial alignment in the interfacial region, the entangled conformation of the grafted molecules and the LCE matrix produced a semi-rigid (condensed) microstructure.

The strain energy density distributed to each component was calculated after the application of uniaxial tensile loading of 0.25% in both the axial and transversal directions. All the components in the system, including the mesogen and side chain in the LCE matrix, PMHS backbone, grafted unit, and AuNP, were independently grouped to characterize the load transfer properties according to each structural feature, as illustrated in Fig. 10(a). The volumes for calculating the virial stress of each group are unified to the entire volume of the unit cell; hence, the sum of the energy density directly represents the mechanical properties of the neat LCE and the LCE/AuNP nanocomposite models.

The analysis results are shown in Fig. 10(b) and **(c)**, and several important observations were made. First, regardless of the loading direction, most of the external force applied to the microstructure was stored in the mesogens, followed by the side chains directly connected to the mesogens. In addition to the neat LCE model, the LCE/AuNP nanocomposite models also showed the same result, except that the grafted units on the AuNP achieved a considerable level of strain energy storage.

Second, the strain energy of both the mesogen and the side chain groups exhibited high anisotropy in the axial and transverse directions. The strain energy of both groups further increased under axial loading as the orientational order of the LCE matrix increased (for the nanocomposite systems with the larger AuNPs), whereas the opposite effect was observed under transverse loading (see Figure S3 in the Supplementary Data). Simultaneously, the strain energy difference between the mesogen unit and the side chain was more pronounced in the transverse direction than in the axial direction. The results provide a principal clue to explain the relationship between the structure and load-transfer properties of the LCE components. The mesogen units have a planar conformation in which the two rigid benzene rings are connected to each other and are compactly arranged in the nematic direction. Therefore, the intermolecular interaction between the mesogen units acts dominantly on the energy storage characteristics under mechanical loads. It also shows good consistency when the non-bonded and the covalently-bonded strain energy of each component were independently derived, as shown in Fig. 10(d). The intermolecular interaction (representing the

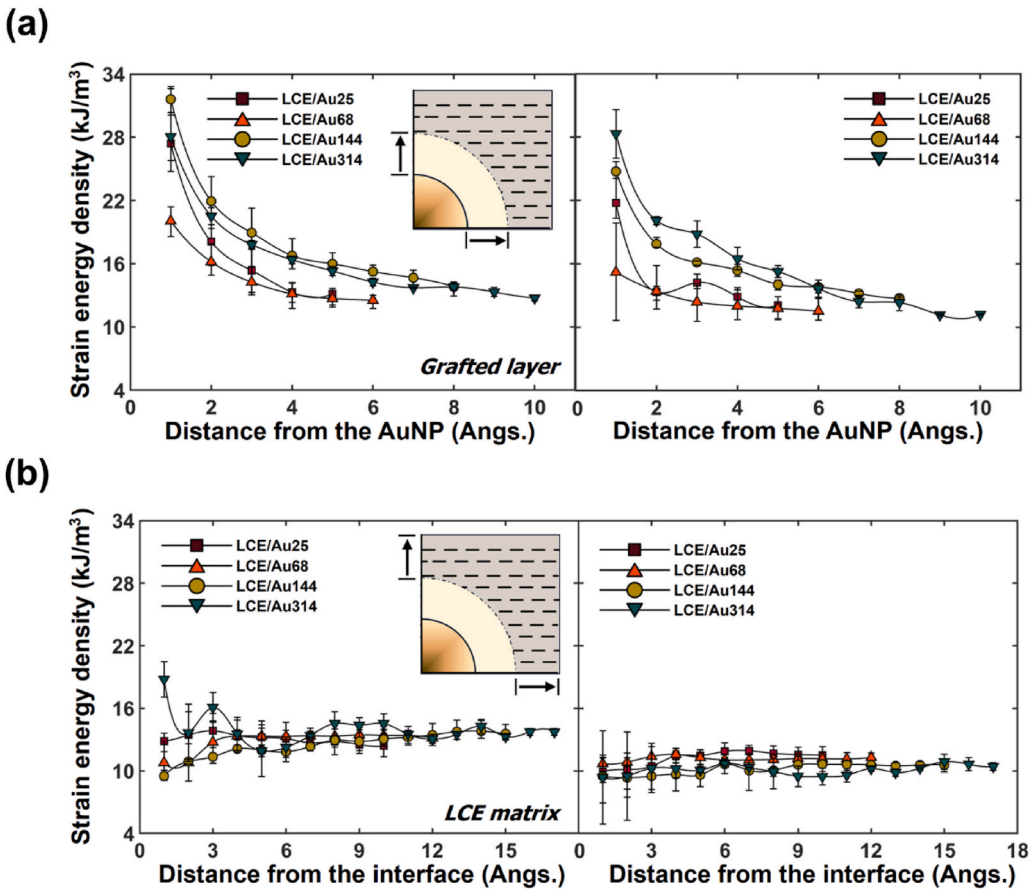



**Fig. 9.** Distributions of strain energy density in the LCE/Au nanocomposites after tensile deformation of 0.25%. The observation range was divided into two regions: (a) the region dominated by grafted molecules and (b) the region mainly composed of the LCE matrix constituents. The tensile deformation was applied along the axial direction (left) and along the transverse direction (right). Error bar represents min-max range.

non-bonded strain energy) of the mesogen is dominant regardless of the deformation direction. Thus, the strain energy of the mesogen units shows only a slight difference in the axial and transverse directions. On the other hand, the mechanical response of the side chain is highly affected not only by the intermolecular interaction, but also by the intramolecular interaction (representing the covalently-bonded strain energy) due to the flexibility of the linear structure. In the LCE matrix, there remains a partial arrangement of the side chains in the nematic director due to the connection with the highly ordered LCE mesogens. Furthermore, the side chain effectively bears the tensile load only in the parallel direction owing to the structural characteristics of the linear chain. From the above factors, the strain energy density of the side chain varies depending on the tensile direction, and a relatively high anisotropy of strain energy is observed. Meanwhile, the strain energy of the PMHS backbone did not significantly change in either direction, regardless of the size of the AuNPs. Owing to the random arrangement and high flexibility inherent in the PMHS backbone, most of the external load applied was converted into intramolecular strain energy.

Third, a considerable amount of strain energy was stored in the grafted molecules of the LCE/AuNP nanocomposite models despite their small volume fraction. The strain energy exhibited a constant behavior in both the axial and transverse directions, regardless of the AuNP size. Although each grafted chain has a linear structure similar to the side chain in the LCE matrix, the alignment radiates from the center of the AuNP, and the grafted chains are highly entangled with the surrounding LCE matrix components. Accordingly, the grafted molecules have a morphological advantage in withstanding external loads in all directions. In addition, the external load transmitted from the LCE matrix was mostly stored as non-bonded strain energy despite the deformability of the linear structure. Because the grafted molecules non-covalently interact with the LCE matrix constituents, the deformation of the surrounding medium almost does not affect the conformational changes. In other words, the mechanical response of the grafted molecules is dominantly represented by the non-bonded interaction, and the intramolecular structure is well-preserved during elongation.

Finally, a small proportion of the external load was transferred to the AuNP owing to the ductility of the grafted layer on the AuNP surface. The result indicated that the grafted molecules absorbed almost all the deformation energy transferred from the LCE matrix and played a 'cushioning role' between the LCE matrix and the AuNP. This result is in good agreement with the consideration exemplified in Fig. 9 in which only the presence of AuNPs among the grafted-AuNPs caused the mechanical degradation in the LCE/AuNP nanocomposite models.

To comprehensively understand the results of the mechanical behavior covered so far, we provided a schematic of the deformation process for the LCE/Au nanocomposites, as shown in Fig. 11. The anisotropic properties of the LCE matrix induced by the arrangement of mesogen and side chains can be represented by different spring constants ($k_{Axial}$ and $k_{Trans}$). Owing to the radial arrangement of the grafted units, the grafted layer preserved the isotropic mechanical behavior, which can be represented by the same spring constant ($k_{Iso}$) in all directions. The present study estimated the interface between the LCE matrix and the grafted layer as perfectly bonded with their interlocked structure, facilitating highly efficient stress transfer. On the other hand, the grafted layer that absorbed all the transferred energy in the immediate vicinity of the AuNP surface was evaluated as a de-bonded interface that maintained the unloaded state of the AuNPs.

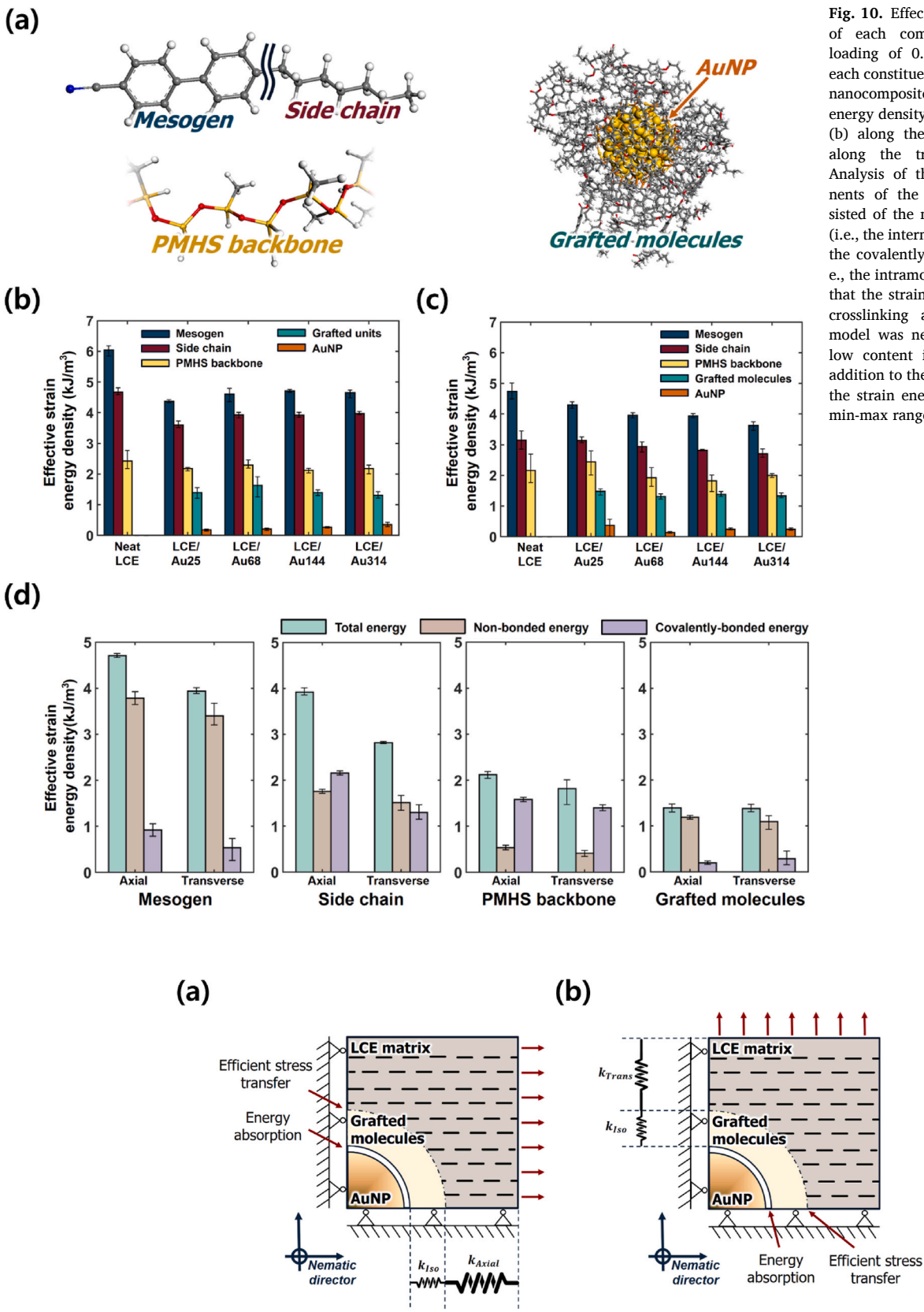



**Fig. 10.** Effective strain energy density of each component due to tensile loading of 0.25%. (a) Description of each constituent of the LCE and LCE/Au nanocomposites. The obtained strain energy density after tensile deformation (b) along the axial direction and (c) along the transverse direction. (d) Analysis of the strain energy components of the LCE/Au144 model consisted of the non-bonded strain energy (i.e., the intermolecular interaction) and the covalently-bonded strain energy (i.e., the intramolecular interaction). Note that the strain energy of the diacrylate crosslinking agent in the neat LCE model was neglected because of their low content in the microstructure in addition to the negligibly small value of the strain energy. Error bar represents min-max range.




**Fig. 11.** Schematic view of the mechanical loading system of the LCE nanocomposites embedded with grafted-AuNP. The tensile loading applied in (a) the axial direction and (b) the transverse direction.

## 4. Conclusions

In this study, the mechanical and interfacial properties of LCE nanocomposites functionalized with grafted-AuNPs were examined through MD simulations. The load transfer characteristics, impacted by the grafted-AuNP according to different particle sizes, as well as the associated inner morphology were quantitatively identified.

The results indicated that the insertion of grafted-AuNPs not only significantly changed the alignment characteristics of the LCE components near the grafted layer, but also had an adverse effect on the mechanical properties of the nanocomposites. The increased effective size of the AuNPs due to the thick grafted layer caused significant misalignment of adjacent mesogen units, and the effect gradually declined over long distances. In particular, as the size of the AuNPs decreases under a constant volume fraction, the recovery of alignment of the LCE components becomes difficult because of the short spatial distance between the grafted-AuNPs. In addition, through the strain energy density analysis assigned to each component, it is elucidated that the grafted layer is in the highly loaded state, whereas the AuNP is in the unloaded state during the tensile deformation.

We reached two important conclusions regarding the role of the grafted nanoparticles in the polymer nanocomposites. First, the interlocked interface between the grafted molecules and the LCE components achieves high structural rigidity that resists deformation and ensures effective stress transfer. Second, the grafted molecules on the AuNPs have high ductility, which easily deforms under external loads and absorbs most of the strain energy, providing a cushioning effect that prevents the transfer of load to the rigid AuNPs. These two findings explain how the chemical surface treatment of nanoparticles during the synthesis of functional nanocomposites critically affects the mechanical behavior of the finished products.

The current MD study is the first step to present a new perspective for the design of LCE nanocomposites that support high loadings of the grafted nanoparticles. Therefore, it is expected that the results obtained in this study will be utilized in practical applications by transferring them to a multiphase finite element model, which enables the comprehension of mechanics at the molecular level as a continuum description. The orientational order and associated anisotropic stiffness of the matrix that varied with the size of the AuNPs can be estimated. In particular, the results of strain energy due to each molecular component provide an illustration of the contact condition at the interface in composites or bonded components.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.polymer.2021.123525.

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